

Laboratory and computational workbook for Labs 01-09, with instructor keys and grading rubrics

Labs 01-09: Simulation, computational and (SOP-governed) teaching-lab formats

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All datasets used in this package are simulated teaching data. Wet-lab and animal procedures described here are conceptual; institutional biosafety approval and SOPs govern all practice.

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Lab 01 — Molecular Cloning Design Exercise

Type: Simulation (no living organisms) · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Workbook](#) · [■ Cloning data](#) · [■ Assessment](#)

1. Learning objectives

By the end of this lab you will be able to:

1. Design a restriction-based cloning plan from a vector map and insert sequence.
2. Choose enzymes for directional cloning and justify the choice.
3. Design PCR primers that add restriction sites and maintain reading frame.
4. Compute insert:vector molar ratios for ligation setup.
5. Predict the diagnostic-digest pattern that will confirm the final construct.

2. Background

Every cloning project begins *in silico*. Wrong design decisions cascade into wasted weeks; right ones make the bench work routine. This exercise simulates the full design phase for a reporter construct you will "build" in the following labs.

Scenario: You study a zebrafish developmental gene, ***shha*** (sonic hedgehog a), and want to test whether its promoter drives expression in the notochord/floor-plate. You will clone a 850 bp *shha* promoter fragment upstream of **GFP** in a 3.2 kb plasmid vector.

3. Principle

Directional restriction cloning: two different restriction enzymes generate non-compatible sticky ends on both vector and insert, forcing a single orientation. The design must verify (a) the chosen sites are unique in the vector, (b) they are absent from the insert, (c) reading frame and any tag junctions are preserved, (d) a diagnostic digest exists that distinguishes correct clones from empty vector.

4. Materials (simulation)

- Vector map `DATA/cloning-data/pDevGFP_map.txt` (MCS: EcoRI, BamHI, HindIII, XhoI — all unique; ori; kanR; T7/SP6 priming sites; lacZ α)
- Insert sequence `DATA/cloning-data/shha_promoter_850bp.txt`
- In-silico tools: any restriction analyzer (or the provided scripts `DATA/cloning-data/design_check.py`)

5. Equipment

Computer with Python 3 or a plasmid-design application; text editor; calculator.

6. Safety

Simulation only — no organisms, no enzymes, no hazards. (When you later perform real digestion/ligation labs, your institution's biosafety and chemical-hygiene SOPs apply.)

7. Procedure / workflow

1. **Inspect the vector map.** Record total size, marker(s), MCS sites, priming sites.
2. **Scan the insert** for EcoRI/BamHI/HindIII/XhoI internal sites (script or manual).
3. **Choose the enzyme pair.** Justify with three criteria: unique in vector, absent in insert, directional.

4. **Design primers** that add the chosen sites (with 4–6 nt 5' clamps for enzyme efficiency) to the 850 bp insert. Write the primer sequences.
5. **Plan the ligation** using the molar-ratio formula (below) for a 50 ng vector amount and 3:1 ratio.
6. **Design the diagnostic digest:** pick one enzyme pair that (a) cuts the final construct into ≥ 2 informative fragments and (b) gives a *different* pattern for empty vector.
7. **Predict everything in silico:** gel bands for (i) undigested construct, (ii) diagnostic digest, (iii) empty-vector digest.

Molar-ratio formula:

$$\text{ng insert} = \text{ng vector} \times (\text{bp insert} / \text{bp vector}) \times (\text{molar ratio desired})$$

8. Expected results

A complete, self-consistent design package: enzyme pair, primer table, ligation calculation, digest prediction table. There is **one best enzyme pair** given the insert's internal sites (the scripts will reveal it) — discovering why the others fail is the point.

9. Data tables to complete

Item	Your answer
Vector size / marker / MCS sites	
Internal-site conflicts found	
Enzyme pair chosen + justification	
Forward primer (5'→3')	
Reverse primer (5'→3')	
Insert ng for 50 ng vector at 3:1	
Diagnostic digest fragments (construct)	
Diagnostic digest fragments (empty vector)	

10. Calculations

- Molar-ratio calculation (show work).
- Total expected construct size (vector + insert).
- Expected fragment sizes for the diagnostic digest.

11. Interpretation

- Why does the chosen pair enforce directionality?
- Which design error would produce *colonies on the plate but no correct clones*?

12. Troubleshooting (design-level)

Symptom	Likely cause	Fix
Enzyme sites not unique	Internal site in insert	Alternative enzymes; Gibson instead
Frame broken at junction	Primer mis-design	Recheck frame/stop codon logic
Digest uninformative (one band)	Poor diagnostic choice	Pick asymmetric sites

13. Post-lab questions

1. Why add 5' clamp bases before restriction sites in primers?

2. Your insert contains one HindIII site. Give two alternative strategies.
3. What colony-PCR primer design would let you distinguish orientation without a digest?
4. Why is it useful to predict the *empty-vector* digest pattern too?

14. Viva questions

1. Define directional cloning.
2. What is the role of the 5' clamp in primer design?
3. Which vector component determines colony selection here?

15. Instructor notes / answer key (summary)

- Correct pair: **EcoRI + HindIII** (BamHI is blocked by an internal site in the insert; XhoI lacks a distal mate for directionality in this design).
- Primer design must add 5'-clamp + EcoRI (forward) and 5'-clamp + HindIII + reverse-complement (reverse).
- 50 ng vector at 3:1 → insert ng = $50 \times (850/4050) \times 3 \approx \mathbf{31.5 \text{ ng}}$.
- Diagnostic digest (EcoRI+BamHI): construct → 1.9 kb + 2.15 kb; empty vector → single 3.2 kb.
- Full worked answers in the Workbook.

16. Advanced challenge

Redesign the same construct with **Gibson Assembly**: write the 30-bp overlaps you would use, and state one scenario where Gibson is clearly the better choice here.

Lab 02 — Restriction Digestion Analysis (Simulation)

Type: Simulation / Computational (uses provided data; optional wet-lab demonstration per institutional SOP)

• **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Workbook](#) · [■ Cloning data](#) · [■ Assessment](#)

1. Learning objectives

1. Relate restriction-enzyme recognition/cut positions to fragment outcomes.
2. Build an in-silico digest and predict fragment sizes.
3. Interpret simulated gel images/migration data against a ladder.
4. Diagnose cloning outcomes (correct, empty, rearranged) from digest patterns.

2. Background

A diagnostic digest is the cheapest structural check of a plasmid: cut with enzymes whose sites flank the insert (or cut asymmetrically), run the fragments on an agarose gel, and compare observed sizes with the map prediction. Because migration $\propto \log(\text{length})$, sizing requires interpolation against a ladder.

3. Principle

- Enzymes cut at defined sequences (recognition sites) at defined positions (see Module 4).
- Fragment sizes depend on *positions* of sites, not just their presence.
- Comparing **predicted** to **observed** sizes validates (or falsifies) the assumed map.

4. Materials (simulation)

- `DATA/cloning-data/plasmid_sequences/*.fasta` — three candidate plasmids A, B, C (one correct, one empty vector, one rearranged)
- `DATA/cloning-data/gel_migration_ladder.csv` — ladder band sizes and migration distances
- `DATA/cloning-data/digest_band_positions.csv` — band positions for each digest lane (A, B, C × two enzyme pairs)
- `DATA/cloning-data/analyze_digest.py` — interpolation script
- Optional wet-lab demonstration: instructor-run digest of a teaching plasmid (follow institutional SOP; standard agarose-gel materials).

5. Equipment

Computer (Python 3); optional: teaching-lab gel apparatus (demonstration only).

6. Safety

Simulation: none. Wet-lab demonstration: follow institutional chemical/biological hygiene SOPs (SYBR-class stains, UV/blue-light handling) under instructor supervision.

7. Procedure / workflow

1. Read the plasmid maps (FASTA + annotation table).
2. Use `analyze_digest.py` (or any restriction tool) to compute fragment sizes for each enzyme pair on each plasmid.
3. Convert band *positions* to *sizes* using the ladder (log-interpolation).

- Match observed to predicted; classify each plasmid as **correct / empty / rearranged**.
- For the rearranged plasmid, propose a molecular explanation consistent with the pattern.

Interpolation snippet:

```
import pandas as pd, numpy as np
lad = pd.read_csv("DATA/cloning-data/gel_migration_ladder.csv")
coef = np.polyfit(lad.migration_mm, np.log10(lad.size_bp), 1)
def size_from_pos(pos): return 10 ** np.polyval(coef, pos)
```

8. Expected results

Lane	Observed sizes (≈)	Diagnosis
A digest 1	4.05 kb + 0.85 kb	correct construct
B digest 1	3.2 kb only	empty vector
C digest 1	2.3 + 1.7 + 0.9 kb	rearranged (e.g., partial duplication/inversion)
A digest 2 / B digest 2 / C digest 2	consistent with diagnoses	

(Exact values in the CSVs; the workbook shows full reasoning.)

9. Data table

Plasmid	Enzyme pair	Predicted sizes	Observed sizes	Diagnosis
A				
B				
C				

10. Calculations

- Ladder interpolation (show the fit and one example conversion).
- Predicted vs observed error (%) for each band.

11. Interpretation

- Why does an *undigested* plasmid not give a single size band? (Supercoiled forms migrate anomalously.)
- Why use two enzyme pairs for a definitive call?

12. Troubleshooting

Symptom	Likely cause	Fix
Smearing lanes	DNA degradation / overload	Clean prep; load less
Faint high-MW band	Partial digest	Fresh enzyme; check star activity
Bands "between" ladder	Normal — interpolate	—

13. Post-lab questions

- Why does a *linearized* empty vector sometimes appear larger than its map size?
- Your correct-construct digest shows three bands, not two. What third site could explain it?
- Why is log(size) linear in migration, not size itself?

14. Viva questions

1. Give one enzyme producing sticky ends and one producing blunt ends.
2. Why compare against a ladder rather than the well position?
3. What is star activity?

15. Instructor notes / answer key (summary)

- A = correct (expected fragments match design), B = empty vector, C = rearranged (duplication of a region — inferable because the sum of observed fragment sizes exceeds the plasmid map size).
- Wet-lab variant: use an instructor-prepared teaching plasmid with a single-insert design; run one-enzyme and two-enzyme digests to show the contrast.
- Full table in the Workbook.

16. Advanced challenge

Design a *single-enzyme* digest strategy that still distinguishes orientation for the Lab-01 construct (hint: asymmetric sites around the insert), and explain when a single digest is insufficient.

Lab 03 — Ligation and Transformation Workflow (Simulation)

Type: Simulation (wet-lab procedures are institution-SOP dependent) · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: ↑ [Course Home](#) · [Workbook](#) · ■ [Cloning data](#) · ■ [Assessment](#)

1. Learning objectives

1. Set up a ligation logically: molar ratios, controls, and dephosphorylation decisions.
2. Predict transformation outcomes from ligation composition.
3. Interpret plate counts (positive/negative/experimental) quantitatively.
4. Decide which colonies to screen and why.

2. Background

Ligation joins prepared insert and vector; transformation introduces the products into competent cells. Both steps are probabilistic — the design of controls is what makes the outcomes interpretable. This lab simulates the workflow so students learn the *decision logic* before (or instead of) bench practice.

3. Principle

- T4 DNA ligase seals phosphodiester bonds between adjacent 5'-phosphate and 3'-OH ends; ATP-dependent.
- Vector self-ligation competes with insert incorporation → dephosphorylation of vector 5' ends (where appropriate) and correct insert:vector molar ratios matter.
- Chemical transformation and electroporation introduce plasmids into competent cells with empirically defined efficiency — *follow institutional SOPs; protocols vary by strain/equipment*.

4. Materials (simulation)

- DATA/cloning-data/ligation_setup.csv — vector/insert masses, sizes, ratios for four ligation conditions
- DATA/cloning-data/transformation_plates.csv — colony counts for each condition
- Calculator or the workbook sheet

Conditions (teaching design):

Ligation	Vector	Insert	Dephosphorylated?	Purpose
L1	+	–	yes	vector-only control (background)
L2	+	+ (3:1)	yes	experimental
L3	+	+ (0:1... see data)	no	ratio-error demonstration
L4	+	+ (10:1)	yes	insert-excess demonstration

5. Equipment

Computer/calculator. (Wet-lab version requires competent cells, selective plates etc. **per institutional SOP and instructor approval only.**)

6. Safety

Simulation: none. Wet-lab version: standard BSL-1 containment and disinfection per institutional SOPs; no operational parameters are specified here by design.

7. Procedure / workflow

1. Compute ng of insert needed for each ratio (Lab-01 formula).
2. Complete the ligation-setup table.
3. Open `transformation_plates.csv`; compute colonies per condition (and colonies/ng if provided).
4. Classify each condition's outcome: background level, good yield, ratio error, or contamination.
5. For the best condition, decide *how many colonies to screen* and which screen (colony PCR vs blue-white) fits this vector (kanR + lacZ α , per Lab-01 map).

8. Expected results

- L1 (vector-only) → low background (a few colonies) — the baseline empty-vector rate.
- L2 (3:1, dephosphorylated) → highest proportion of correct clones.
- L3 → high empty-vector background (no dephosphorylation).
- L4 → many multi-insert/concatemer events (higher-than-expected insert-size colonies).

9. Data table

Condition	Colonies	Empty-vector rate	Correct-clone rate (from later screening data)
L1			—
L2			
L3			
L4			

10. Calculations

- Insert ng per condition (3:1 and 10:1 with 50 ng vector; insert 850 bp, vector 3.2 kb).
- Transformation efficiency = colonies / μg DNA (where the dataset provides it).

11. Interpretation

- Why does dephosphorylation *reduce* L1 colonies specifically?
- Why does excess insert produce concatemers rather than more correct clones?

12. Troubleshooting (conceptual)

Symptom	Likely cause	Fix
No colonies anywhere	Dead competent cells / wrong antibiotic	Positive-control transformation
L2 $\approx$ L1	Ligation failed or insert ends incompatible	Recheck digest/ends
Too many colonies on L1	Over-dephosphorylation failure; over-plating	Repeat with fresh CIP/rSAP step

13. Post-lab questions

1. Why is the L1 control essential to interpreting L2?

2. What would you change first if L4 yields concatemers?
3. Why must the "correct-clone rate" be measured by screening, not by colony number?

14. Viva questions

1. What cofactor does T4 ligase require?
2. Define competent cells (conceptual).
3. Why dephosphorylate the vector in blunt-end cloning *a/ways*, but not necessarily in two-enzyme directional cloning?

15. Instructor notes / answer key (summary)

- Expected pattern: $L1 < L3 < L2$ in correct-clone rate; L4 shows concatemers. Colony counts are in the CSV; class-average discussion points are in the Workbook.
- Wet-lab conversion: if the institution permits a real transformation teaching exercise, substitute the approved SOP for Section 7 and add the SOP reference to Section 6; keep the same data tables.

16. Advanced challenge

Design a ligation strategy using **two inserts** (promoter + GFP) into one vector in a *single* reaction — state the order/orientation logic and which technique (restriction vs Gibson vs Golden Gate) you would pick and why.

Lab 04 — Colony PCR and Clone Screening (Simulation)

Type: Simulation / Computational · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Workbook](#) · [■ Cloning data](#) · [■ Assessment](#)

1. Learning objectives

1. Design colony-PCR primers that distinguish empty vs insert-carrying vs wrong-orientation clones.
2. Interpret simulated colony-PCR band patterns.
3. Choose and interpret a diagnostic digest for the PCR-positive clones.
4. Explain what Sanger sequencing adds and design primer coverage across the construct.

2. Background

After transformation, *selection* (antibiotic) only says "has vector". *Screening* (colony PCR, blue-white) shortlists candidates; *validation* (digest, sequencing) proves structure. This lab walks that decision chain on simulated colonies from Lab 03's L2 condition.

3. Principle

- **Colony PCR:** primers flanking the MCS give a size-discriminating product; one vector + one insert primer forces orientation-specific readout.
- **Blue-white (this vector carries lacZ α):** white colonies = insert-disrupted lacZ α candidates; blue = empty vector.
- **Diagnostic digest:** structural check of mini-prep DNA (Lab 02 logic).
- **Sanger sequencing:** junction + internal coverage; final word on sequence identity.

4. Materials (simulation)

- DATA/cloning-data/colony_pcr_results.csv — 24 colonies × band sizes (two primer sets)
- DATA/cloning-data/sanger_chromatogram_summary.txt — synthetic trace descriptions (good, mixed, truncated)
- Primer sequences from Lab-01 design

5. Equipment

Computer (Python/pandas optional for tabulating patterns).

6. Safety

Simulation only.

7. Procedure / workflow

1. **Design two primer sets:**
 - Set A: vector-flanking (both primers in vector, outside MCS) — size distinguishes empty vs insert; cannot tell orientation.
 - Set B: vector-forward + insert-reverse — orientation-specific.
2. **Tabulate the 24 colonies** from the CSV: classify each as empty / correct / wrong-orientation / ambiguous / PCR-failure.

3. **Cross-check with the blue-white data** (the CSV includes the plate color): white+PCR-positive → candidate; blue+PCR-positive → beware (lacZ α not disrupted — internal-site anomaly?); white+PCR-negative → re-run PCR.
4. **Choose mini-prep clones** (top 4) and predict their diagnostic-digest patterns.
5. **Plan Sanger coverage**: given vector primers bind 120 bp from the MCS, how many reads cover the 850 bp insert + junctions? Propose an internal primer if needed.

8. Expected results

Pattern (Set A)	Pattern (Set B)	Diagnosis
~1.05 kb	~0.7 kb	correct orientation
~1.05 kb	none	wrong orientation
~0.2 kb	none	empty vector
none	none	PCR failure (re-run)
both A bands	mixed	mixed colony / indel — re-streak

(Simulated data will contain ~60–70% correct, plus deliberate failure modes.)

9. Data table

Colony	Plate color	Set A band	Set B band	Diagnosis	Next step
1					
...					

10. Calculations

- Expected band sizes for Sets A and B from your Lab-01 map.
- Read coverage arithmetic for Sanger (read length ~700–900 bp assumption, stated).

11. Interpretation

- Why can Set A never resolve orientation?
- Why must a blue colony with a positive PCR be investigated rather than discarded silently?

12. Troubleshooting

Symptom	Likely cause	Fix
All colonies PCR-negative	PCR setup error; too little colony template	Positive-control colony; fresh mix
Both bands in Set A	Mixed colony	Restreak single colony
Correct PCR but odd digest	PCR can't see structural changes	Trust the digest; sequence

13. Post-lab questions

1. Why pick both white *and* blue colonies when troubleshooting the plate, rather than white only?
2. How many Sanger reads would definitively cover your 4.05 kb construct, and why?
3. A colony is PCR-positive in Set B but digest-negative. Which do you trust, and why?

14. Viva questions

1. Define screening vs selection (Module 3).

2. Why does lacZ α complementation require a specific host strain?
3. What does a mixed Sanger trace (overlapping peaks) indicate?

15. Instructor notes / answer key (summary)

- The CSV deliberately includes: 14 correct, 4 empty, 2 wrong-orientation, 2 mixed, 2 PCR-failure colonies.
- Sanger coverage: two vector-primed reads (F and R) cover insert + junctions for this 850 bp insert; anything >1 kb needs an internal primer.
- Full walk-through in the Workbook.

16. Advanced challenge

Design a screening pipeline that avoids Sanger sequencing entirely (restriction + PCR logic only), and state explicitly what residual risk it leaves (e.g., PCR-derived point mutations in the insert).

Lab 05 — Gene-Expression Analysis with qPCR Data

Type: Computational (simulated qPCR dataset) · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Workbook](#) · [■ Gene-expression data](#) · [■ Assessment](#)

1. Learning objectives

1. Apply the $\Delta\Delta\text{Ct}$ method to a multi-replicate qPCR dataset.
2. Choose and justify reference genes using provided stability data.
3. Interpret a developmental time course (zebrafish *shha*, *myod1*, *pax6a*).
4. Report results to course standards (n, normalization, statistics, effect size).

2. Background

RT-qPCR quantifies transcripts relative to references. The $\Delta\Delta\text{Ct}$ method converts Cq values to fold changes; the analysis is only as good as (a) RNA quality, (b) primer specificity/efficiency, and (c) reference-gene stability in the *actual* comparison. This lab uses a simulated zebrafish developmental time course (10–24 hpf, 4 stages × 4 biological replicates × 3 genes + 2 reference candidates).

3. Principle

$$\Delta\text{Ct} = \text{Cq}(\text{target}) - \text{Cq}(\text{reference})$$

$$\Delta\Delta\text{Ct} = \Delta\text{Ct}(\text{stage}) - \Delta\text{Ct}(\text{calibrator stage})$$

$$\text{FC} = 2^{(-\Delta\Delta\text{Ct})}$$

Assumes ~100% amplification efficiency; efficiency-corrected methods (Pfaffl) when measured efficiencies differ. Statistics are performed on ΔCt (or $\log_2 \text{FC}$), not on fold changes.

4. Materials

- DATA/gene-expression-data/qPCR_timecourse.csv — Cq values (targets: *shha*, *myod1*, *pax6a*; references: *ef1a*, *rpl13a*; stages: 10, 12, 18, 24 hpf; n = 4 replicates)
- DATA/gene-expression-data/reference_gene_stability.csv — computed M-values/geNorm-style stability
- Python 3 (pandas, matplotlib) or spreadsheet software

5. Equipment

Computer with Python 3 and pandas/matplotlib.

6. Safety

Simulation only.

7. Procedure / workflow

1. Load the CSV; compute mean Cq per (gene, stage, replicate).
2. Evaluate reference stability from the provided table; choose the stable reference (or the geometric mean of both).
3. Compute ΔCt , then $\Delta\Delta\text{Ct}$ vs the 10 hpf calibrator, then fold change.
4. Plot $\log_2$ fold change vs stage for the three genes.
5. Test stage effects on ΔCt (ANOVA or linear model — your choice; state it).

6. Write a results paragraph to course reporting standards.

8. Expected results (shape of the story)

- ***shha***: expression present from 10 hpf, modest change — early patterning gene.
- ***myod1***: strong rise between 12→18 hpf — myogenic commitment wave.
- ***pax6a***: rise from 12 hpf and plateau — anterior/neural specification.
- (*Simulated magnitudes; interpret the pattern, not the exact values.*)

9. Data table

Gene	10 hpf (calibrator)	12 hpf FC	18 hpf FC	24 hpf FC	Stage effect p
<i>shha</i>	1.0				
<i>myod1</i>	1.0				
<i>pax6a</i>	1.0				

10. Calculations

- Full $\Delta\text{Ct}/\Delta\Delta\text{Ct}/\text{FC}$ worked for *myod1* at 18 hpf (show arithmetic on replicate means of ΔCt).
- Reference-gene choice justification from stability values.

11. Interpretation

- Why analyze ΔCt statistically rather than fold change?
- Why is 10 hpf a reasonable calibrator here, and what would change if you picked 24 hpf?
- How would you *validate* the *myod1* result at the spatial level? (Method choice from Module 11.)

12. Troubleshooting (conceptual)

Symptom	Likely cause	Fix
One replicate Cq 3 cycles late	Pipetting/RNA-quality outlier	Investigate; document exclusion rule
No-RT control amplifies	Genomic contamination	DNase; intron-spanning primers
References shift with stage	Unstable reference	Use validated alternative(s)

13. Post-lab questions

1. Why is $n = 4$ *biological* replicates the minimum here rather than 4 technical replicates?
2. Your no-template control shows amplification at Cq 35. Diagnose.
3. Which spatial method would you choose to confirm the *myod1* result, and what would a positive result look like?

14. Viva questions

1. Define Cq/Ct.
2. Why must reference genes be validated per experiment?
3. What does a melt-curve peak at the wrong temperature indicate?

15. Instructor notes / answer key (summary)

- Reference choice: *ef1a* and *rpl13a* both stable; geometric-mean normalization is best practice.

- *myod1* 18 hpf: $\Delta Ct \approx 8.6 \rightarrow \Delta\Delta Ct \approx -2.9 \rightarrow FC \approx 7.5\times$ vs 10 hpf (simulated values; workbook shows the arithmetic).
- Full solution and plots in the Workbook.

16. Advanced challenge

Add an efficiency-correction step: given per-primer efficiencies from a (provided) dilution series table, recompute fold changes with the Pfaffl method and compare conclusions with $\Delta\Delta Ct$.

Lab 06 — Reporter-Gene Analysis (Simulation + optional demonstration)

Type: Simulation (fluorescence/luminescence datasets); optional wet demonstration per SOP · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: ↑ [Course Home](#) · [Workbook](#) · ■ [Gene-expression data](#) · ■ [Assessment](#)

1. Learning objectives

1. Design promoter- and enhancer-reporter experiments with proper controls.
2. Interpret dual-luciferase data (Firefly normalized to Renilla).
3. Interpret fluorescence-microscopy reporter images (simulated).
4. Distinguish reporter output from endogenous-expression claims (validity limits).

2. Background

Reporters convert regulatory activity into measurable signal (Module 5). The design principles: promoter-reporters ask *when/where is this promoter active*; enhancer-reporters ask *does this sequence enhance, and where*; fusions ask *where does the protein go*. Each design has characteristic failure modes that controls are built to catch.

3. Principle

- **Dual luciferase:** Firefly luciferase under the test regulatory element; Renilla under a constitutive promoter as transfection/internal control; Firefly/Renilla ratio = normalized reporter activity.
- **Fluorescence reporters:** GFP intensity per cell/tissue region reports regulatory activity qualitatively—semiquantitatively; background subtraction and exposure control are essential.

4. Materials (simulation)

- DATA/gene-expression-data/dual_luciferase.csv — Firefly/Renilla raw values: test promoter (WT), test promoter (mutated motif), minimal-promoter-only, enhancer ± orientation, empty vector; n = 4
- DATA/gene-expression-data/gfp_reporter_images_summary.csv — simulated image-region intensities (notochord, somite, neural tube) for promoter-GFP constructs
- Python 3 or spreadsheet

5. Equipment

Computer (Python 3: pandas, matplotlib).

6. Safety

Simulation only. (A wet demonstration of luciferase assay readout in a plate reader may be run by instructors per institutional SOP — no student-operated transfection in the teaching setting.)

7. Procedure / workflow

1. **Compute normalized activities:** Firefly/Renilla per replicate; mean ± SD per construct.
2. **Statistical comparison:** WT promoter vs mutated motif; WT vs minimal-promoter-only (state test and n).

3. Enhancer orientation analysis: A-vs-B orientation of the enhancer — enhancers are orientation-flexible; interpret the result accordingly.

4. Fluorescence region analysis: background-subtract; compare GFP in notochord vs somite vs neural tube regions for the *shha*-promoter-GFP construct.

5. Write the interpretation: does the data support "the promoter drives notochord expression"? What is NOT proven?

8. Expected results (shape)

- WT promoter >> mutated motif (motif-dependent activity).
- WT >> minimal-promoter-only (promoter carries specific activity).
- Enhancer works in both orientations (within error) — the signature of enhancer behavior.
- GFP signal concentrated in the notochord region, matching the endogenous *shha* literature pattern.

9. Data table

Construct	Normalized activity (mean $\pm$ SD)	vs WT
Promoter WT		—
Promoter motif-mutant		
Minimal promoter only		
Enhancer + minimal (orient A)		
Enhancer + minimal (orient B)		
Empty vector		

10. Calculations

- Normalization arithmetic for one full row (raw $\rightarrow$ ratio $\rightarrow$ mean $\pm$ SD).
- Fold-activity WT vs minimal promoter.

11. Interpretation

- Why is Renilla normalization essential in transient transfection?
- What does the enhancer-orientation result *mechanistically* suggest?
- Why does promoter-GFP activity not prove the endogenous gene's spatial pattern? (Chromatin context, missing distal elements — Module 5/10.)

12. Troubleshooting (conceptual)

Symptom	Likely cause	Fix
Renilla counts ~0	Transfection failure or wrong substrate	Check co-transfection controls
Firefly saturated	Over-expression / exposure too long	Dilute lysate; shorten window
High empty-vector activity	Promoterless vector still active (cryptic element)	Re-design; use truly minimal backbone

13. Post-lab questions

1. Design the minimal control set for a *new* enhancer candidate.
2. Your enhancer works in orientation A only. Propose a molecular explanation and a follow-up.
3. When would you add a destabilization tag to GFP, and what question does that change?

14. Viva questions

1. Compare fluorescence vs luminescence quantitation (Module 5 table).
2. Why mutate the motif rather than delete it?
3. Name one fusion-reporter artifact.

15. Instructor notes / answer key (summary)

- Motif mutation reduces activity ~80%; enhancer works both orientations (simulated data in CSV).
- Wet demonstration optional: instructor-run luciferase readout; students interpret provided raw counts.
- Full solution in the Workbook.

16. Advanced challenge

Propose an experimental series that moves from "reporter active in notochord" to "enhancer *necessary* for endogenous expression" (CRISPRi or knockout of the endogenous element; ISH readout). State what each step adds.

Lab 07 — Spatial Gene-Expression Interpretation (In-situ-style images/data)

Type: Simulation / Image-interpretation · **Duration:** 1 session (~3 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Workbook](#) · [■ Spatial data](#) · [■ Assessment](#)

1. Learning objectives

1. Interpret simulated in-situ-hybridization images (domains, gradients, staging artifacts).
2. Design the probe/controls for a new spatial target.
3. Quantify spatial patterns (domain area, intensity profile) from provided data.
4. Distinguish gradient from domain quantitatively.

2. Background

In situ hybridization maps endogenous mRNA spatially (Module 11). Reading ISH images is a core developmental-biology skill: identifying domains, comparing across stages, and recognizing artifacts (staining heterogeneity, probe trapping, staging mismatch).

3. Principle

Antisense probe hybridizes to target RNA; detection yields spatial signal. Sense/no-probe controls bound background; a known marker validates staging and protocol. Signal intensity is *semi*-quantitative — pattern (where) is trustworthy; absolute amount is not.

4. Materials (simulation)

- DATA/spatial-expression-data/ish_sim_images/ — 6 synthetic embryo panels (anterior up): *shha* (notochord/floor-plate domain), *myod1* (somites), *pax6a* (forebrain/eye), *ntl* (presomitic mesoderm + notochord), sense-probe control, no-probe control
- DATA/spatial-expression-data/intensity_profiles.csv — dorsal–ventral intensity profiles per panel
- Python 3 (matplotlib) or image-analysis software
- *All images are generated simulations labeled as such — no real embryos are depicted.*

5. Equipment

Computer (Python 3: matplotlib, pandas).

6. Safety

Simulation only. (Real ISH teaching demonstrations involve fixed specimens and formamide-containing buffers — instructor-demonstration only, per institutional chemical-hygiene SOPs.)

7. Procedure / workflow

1. **Inventory the panels:** gene, stage, view, probe type (antisense/sense/none).
2. **Score each panel** (0 = absent, 1 = weak, 2 = moderate, 3 = strong) by region: notochord, floor plate, somites, PSM, forebrain, eye, neural tube.
3. **Compare antisense vs controls** — establish background level.

4. **Domain vs gradient:** for *shha*, plot the DV intensity profile; fit a monotonic decrease (gradient) vs step (domain) — which does the data support?
5. **Staging check:** use *ntl* (PSM marker) to verify the stage annotation of each panel.
6. **Design the next probe:** pick one gene from Module 8 (e.g., *krox20/egr2b*) and specify probe region, controls, expected pattern.

8. Expected results (shape)

Gene	Domain(s)	Pattern type
<i>shha</i>	notochord/floor plate	graded dorsal decrease
<i>myod1</i>	somites (formed), adaxial cells	domains, stage-dependent
<i>pax6a</i>	forebrain/eye field	broad domain
<i>ntl</i>	PSM + notochord	domains; staging marker
sense / no-probe	uniform faint/no signal	background

9. Data table

Panel	Region scores (0–3 each)	Background level	Pattern type
<i>shha</i>			
<i>myod1</i>			
<i>pax6a</i>			
<i>ntl</i>			
sense			
no-probe			

10. Calculations

- From `intensity_profiles.csv`: compute the dorsal-to-ventral intensity ratio for *shha*; state whether it is consistent with a gradient (continuous) or a domain (step).
- Domain area: threshold the *pax6a* panel at background + 2 SD; compute fraction of embryo area positive.

11. Interpretation

- Why is the sense-probe panel the *only* valid background reference for this protocol?
- A colleague claims *shha* shows a "step function". What plot settles it?
- Why does *ntl* expression in both PSM and notochord complicate staging calls if used alone?

12. Troubleshooting (image-level)

Symptom	Likely cause	Fix
Speckled background	Probe trapping / poor washing	Re-wash; longer stringency
Uniform signal everywhere	Probe too short/repetitive; overstaining	Redesign probe; shorten development
No signal anywhere	Probe degradation; RNA destroyed	Positive-control probe run

13. Post-lab questions

1. Why validate staging with an independent marker rather than morphology alone?

2. Your target gene's ISH shows signal in "everything". Name three causes and fixes.
3. What does ISH *not* tell you about gene expression that a reporter would? (Dynamics.)

14. Viva questions

1. Antisense vs sense probe — define.
2. Why is ISH semi-quantitative at best?
3. What does an RNAscope-class method add over classical ISH?

15. Instructor notes / answer key (summary)

- Scoring keys and profile analysis are in the Workbook.
- The *shha* DV profile in the simulation decreases continuously — gradient-consistent; students should articulate the quantitative criterion used.

16. Advanced challenge

Given the simulated *myod1* panels at two stages, quantify the change in somite-domain area and propose a biological explanation involving somitogenesis timing (Module 9).

Lab 08 — Computational Gene-Expression Analysis (RNA-seq / single-cell / spatial)

Type: Computational · **Duration:** 1–2 sessions (~4–6 h)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: ↑ [Course Home](#) · [Workbook](#) · ■ [Gene-expression data](#) · [Spatial data](#) · ■ [Assessment](#)

1. Learning objectives

1. Run the complete mini-workflows: bulk differential expression, single-cell clustering/annotation, spatial mapping.
2. Apply QC, normalization, dimensionality reduction, and clustering appropriately.
3. Annotate clusters with marker genes and interpret biological identity.
4. Produce publication-style figures (heatmap, PCA/UMAP-style plot, spatial map).

2. Background

Modern expression analysis is computational: count matrices in, biological conclusions out. This lab uses **small simulated datasets** (fully generated, labeled as teaching data) so every step is inspectable — the logic transfers to real pipelines (DESeq2/Seurat/Scanpy class) which students will meet in research settings.

3. Principle

- **Bulk:** normalize for library size → model counts (here: log-CPM + t-test; research: negative-binomial GLM) → FDR.
- **Single-cell:** QC (library size, gene count, doublets) → normalize → PCA → graph/k-means clustering → marker-gene annotation.
- **Spatial:** spot matrix + coordinates → region annotation → region-specific markers → gradient tests.

4. Materials

File	Content
DATA/gene-expression-data/rnaseq_counts.csv	500 genes × 6 samples (WT/KO × 3)
DATA/gene-expression-data/metadata.csv	sample sheet (condition, batch)
DATA/gene-expression-data/single_cell_counts.csv	300 genes × 600 cells (3 latent types)
DATA/gene-expression-data/cell_metadata.csv	cell QC + latent label (instructor use)
DATA/spatial-expression-data/spatial_matrix.csv	200 spots × 200 genes
DATA/spatial-expression-data/spot_coordinates.csv	x,y per spot
DATA/spatial-expression-data/tissue_annotation.csv	spot → region (instructor key)
DATA/spatial-expression-data/analysis_demo.py	starter script

5. Equipment

Python 3 with numpy/pandas/matplotlib/scikit-learn (course venv).

6. Safety

Computational only.

7. Procedure / workflow

Part A — bulk RNA-seq:

1. Load counts + metadata; check batch × condition balance.
2. Filter low-count genes; log-CPM.
3. Compute log2FC (KO vs WT); t-test on log values; BH-FDR.
4. Volcano plot; list top 10 up/down.
5. Interpret: which genes are *plausibly* connected to the KO (provided gene list)?

Part B — single-cell:

1. QC: filter cells with library <500 UMIs or <200 detected genes.
2. Normalize (CPM) + log; PCA (top 2–10 PCs).
3. Cluster (k-means k=3 or graph-based); visualize on PC1/PC2.
4. Annotate clusters using markers (*pax6a* neural, *myod1* muscle, *hbbe1* blood).
5. Compute per-cluster mean expression of a developmental TF set; interpret trajectories.

Part C — spatial:

1. Join matrix + coordinates + annotation.
2. Region-mean expression; top marker per region.
3. Plot each marker across x,y (in-silico in situ).
4. Gradient test: correlate expression with distance-to-edge for one region.

8. Expected results (shape)

- Part A: KO down-regulates ~15–25 genes including a connected pathway; volcano plot separates them.
- Part B: three clusters annotate cleanly as neural/muscle/blood; one subcluster shows intermediate state (transition).
- Part C: markers localize to their regions; one gene shows a monotonic gradient toward the tissue edge.

9. Data tables

Analysis	Metric	Result		
Bulk	# genes FDR<0.05, \	log2FC\	>1	
Single-cell	cluster sizes; marker max-z			
Spatial	top marker per region			

10. Calculations

- Show the log2FC computation for the top gene.
- Show the QC filter arithmetic (cells kept/total).
- Show the gradient correlation (Pearson r, p).

11. Interpretation

- Why must batch not confound condition in Part A?
- What does the intermediate subcluster in Part B suggest biologically, and how would you test it?

- Why is the Part C gradient *hypothesis-generating* rather than conclusive?

12. Troubleshooting (computational)

Symptom	Likely cause	Fix
PCA shows batch separation	Batch effect dominates	Check design; integrate/linear-model
One giant cluster	Over-aggressive QC or wrong k	Re-examine filters; try k via silhouette
Spatial map noise	Spots with tiny libraries	Filter low-library spots

13. Post-lab questions

1. Why is dropout a bigger interpretive problem in scRNA-seq than bulk?
2. What claim can Part C make that Part B cannot — and vice versa?
3. Design the ISH experiment that would validate your Part C top gradient gene.

14. Viva questions

1. Define UMI and its purpose.
2. Why log-transform CPM before PCA/clustering?
3. What is deconvolution in array-based spatial data?

15. Instructor notes / answer key (summary)

- Part A expected: 18 genes $FDR < 0.05$ & $|\log_2 FC| > 1$ (simulated).
- Part B: clusters 1/2/3 $\approx$ neural/muscle/blood with sizes $\sim 220/210/170$; transition subcluster detectable at $k=4$.
- Part C: region markers as designed in the generator script; edge-gradient gene $r \approx -0.6$ (edge-high).
- Starter script + keys: `analysis_demo.py`; full solutions in the Workbook.

16. Advanced challenge

Implement a simple RNA-velocity-*style* directionality argument: given spliced/unspliced-like columns in the single-cell data, order clusters and state the developmental trajectory. Note the assumptions your mini-version makes.

Lab 09 — Capstone Project: From Construct to Expression Pattern

Type: Integrated project (design + analysis + interpretation) · **Duration:** 2–3 weeks (checkpoints weekly)

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: ↑ [Course Home](#) · [Workbook](#) · ■ [All datasets](#) · ■ [Case studies](#)

1. Project statement

Integrate the entire course: **design a construct** → **simulate its cloning and validation** → **design the expression experiment** → **analyze the (simulated) expression data** → **interpret spatiotemporally** → **defend the work**.

Students choose ONE of three tracks:

Track	Focus	Deliverables lean toward
A — Construct engineering	Cloning design depth	Full design dossier + validation plan
B — Expression analysis	Data-analysis depth	Full analysis notebook + figures
C — GMO case synthesis	Applied/regulatory depth	Case dossier + risk/ethics analysis

All tracks share a common spine (below) so grading is comparable.

2. Learning objectives

1. Integrate construct design, cloning, screening, and expression-analysis skills into one coherent project.
2. Apply course reporting standards (n, normalization, statistics, effect size, validation).
3. Defend design decisions in a viva-style presentation.
4. Connect technologies to a real biological or applied question.

3. Common spine (all tracks)

Week 1: Question + design
 ■■■ Track A: construct map, cloning plan, diagnostic predictions
 ■■■ Track B: analysis plan for one provided dataset (which questions, which stats)
 ■■■ Track C: GMO case selection + evidence inventory
 Week 2: Execution (simulated) + data
 ■■■ Track A: in-silico assembly + screening simulation (Labs 1-4 tools)
 ■■■ Track B: run the pipeline; QC + figures
 ■■■ Track C: mechanism + data summary + limitation analysis
 Week 3: Synthesis + defense
 ■■■ Report (≤ 8 pages) + presentation (10 min) + viva

4. Materials

All course datasets and tools; Modules 1–15; prior labs.

5. Equipment

Computer; Python venv; (no wet-lab component — the capstone is design/analysis/interpretation).

6. Safety

No organisms used. Track C must follow the Module 14 layering (evidence → risk → ethics → policy).

7. Procedure / workflow

Track A — Construct engineering (example: conditional reporter)

1. Choose a developmental gene and a spatial question.
2. Design TWO constructs: (i) promoter-GFP reporter; (ii) CRISPRi-ready enhancer-test vector.
3. Full cloning plan: technique choice justified (Module 4 decision guide), primers, ratios, digests, screening plan, sequencing coverage.
4. Predict every failure mode and its detection method.
5. Write the *experiment-day protocol sketch* (conceptual, no operational parameters).

Track B — Expression analysis (example: developmental time course)

1. Pick one dataset (time course, single-cell, or spatial).
2. Pre-register: question, hypotheses, QC thresholds, stats plan (before analyzing!).
3. Execute; produce ≥ 3 figures (profile, PCA/clusters, heatmap or spatial map).
4. Validate-plan: which result needs ISH/reporter follow-up, and how?
5. State limitations explicitly (dropout, resolution, n).

Track C — GMO case synthesis

1. Choose a real GMO case (course list or approved alternative).
2. Reconstruct: modification type, construct logic, mechanism, evidence for benefit, documented risks, regulatory status (**verify current status — cite sources**).
3. Apply the Module 14 four-layer analysis.
4. Propose the *next experiment* that would resolve the case's main open question.

8. Expected results

- Report ≤ 8 pages (or notebook + summary), figures with captions, references (course standard: no fabricated citations).
- Presentation 10 min + 5 min questions.

9. Data table (report template)

Section	Content required
Question	1 paragraph, falsifiable
Design	maps/plan/statistics plan
Results	figures + captions
Interpretation	what is and is NOT proven
Validation plan	next experiments
References	verifiable sources only

10. Calculations

Track-dependent (ratios, digest predictions, $FC/\Delta\Delta Ct$, FDR counts, gradient r).

11. Interpretation

Every track ends with the same question: **what claim does your evidence support, at what resolution, and what is the minimal next experiment?**

12. Troubleshooting (project-level)

Symptom	Likely cause	Fix
Question too broad	Scope creep	One gene, one tissue, one question
Figures without captions	Reporting-standard gap	Caption = what/where/n
Overclaiming	Interpretation gap	Explicit "not proven" list

13. Post-project questions

1. Which single course module was most load-bearing for your track, and why?
2. Where did your original plan change, and what did that teach about design?
3. State one thing your data *cannot* say (resolution or validation limit).

14. Viva questions (examples)

1. Defend your enzyme/technique choice against the alternatives.
2. Why is your reference gene / normalization choice valid here?
3. Your reporter disagrees with endogenous ISH — reconcile.
4. Which regulatory framework applies to your Track-C case in TWO different jurisdictions?

15. Instructor notes / answer key (summary)

- Rubric (100 pts): Question & design 25; Execution/analysis 25; Interpretation & validity 25; Defense & synthesis 15; Reporting standards 10.
- Model dossiers for one example per track are in the Workbook.

16. Advanced challenge

Design the **F0 experiment** that would combine your Track-A construct with a Track-B readout in zebrafish: which reagent goes into which tissue at which stage (conceptually), what controls, and what result would falsify your hypothesis?

GMO, Cloning & Gene-Expression Lab Workbook

Consolidated instructor/student workbook for Labs 01–09

Course: GMOs, Cloning Techniques, and Developmental & Spatial Gene Expression

Navigation: [↑ Course Home](#) · [Labs](#) · [Cheat Sheet](#) · [FAQ](#) · [Answer keys](#)

How to use this workbook: each section pairs one lab with its model answers, worked calculations, and grading notes. Lab documents themselves contain the objectives/procedure; this workbook carries the *solutions and teaching guidance*. Student-facing copies should be issued **without** Sections marked **[Instructor]**.

Contents

1. Lab 01 — Cloning design (solutions)
2. Lab 02 — Restriction digestion (solutions)
3. Lab 03 — Ligation & transformation (solutions)
4. Lab 04 — Colony PCR & screening (solutions)
5. Lab 05 — qPCR expression analysis (solutions)
6. Lab 06 — Reporter analysis (solutions)
7. Lab 07 — Spatial expression (solutions)
8. Lab 08 — Computational expression (solutions)
9. Lab 09 — Capstone (rubric + models)
10. Grading rubrics and reporting standards

Lab 01 — Cloning design

[Instructor] Model answers

- **Vector inventory:** 3,200 bp; kanR; ori high-copy (pUC-class); MCS sites EcoRI, BamHI, HindIII, XhoI all unique; T7/SP6 priming sites flank MCS; lacZ α present.
- **Insert scan:** *shha* promoter 850 bp contains **one internal BamHI** site (position 512); no EcoRI/HindIII/XhoI.
- **Enzyme pair: EcoRI + HindIII.** BamHI fails (internal site); XhoI alone can't give directionality with only one flanking mate in the MCS layout.
- **Primers (model):**
 - Forward: 5'-GCACGAATTC<21 nt of promoter start>-3' (clamp + EcoRI)
 - Reverse: 5'-GCACAAGCTT<revcomp of promoter end>-3' (clamp + HindIII)
 - (Clamps: 4–6 nt; GC clamp at 3' end of the gene-specific portion.)
- **Ligation math (3:1, 50 ng vector):** ng insert = $50 \times (850/4050) \times 3 \approx 31.5$ ng.
- **Diagnostic digest (EcoRI + BamHI):** construct $\rightarrow$ **1.9 kb + 2.15 kb**; empty vector $\rightarrow$ **3.2 kb single band** (BamHI unique; insert adds the second EcoRI-proximal cut point... full derivation in dataset README).
- **Common failure to discuss:** primer design forgetting the clamp $\rightarrow$ enzyme can't cut the PCR product efficiently $\rightarrow$ no insert preparation at all.

Lab 02 — Restriction digestion

[Instructor] Model answers

- Ladder interpolation: fit $\log_{10}(\text{size})$ vs migration $\rightarrow$ linear ($r^2 > 0.99$ in provided data).
- **Plasmid A:** digest-1 observed 4.05 + 0.85 kb $\Rightarrow$ **correct construct** (vector 3.2 + insert 0.85); digest-2 consistent.
- **Plasmid B:** digest-1 observed 3.2 kb only $\Rightarrow$ **empty vector** (single site — the "insert" never integrated).
- **Plasmid C:** digest-1 observed 2.3 + 1.7 + 0.9 kb; **sum = 4.9 kb $\neq$ 3.2 kb** $\Rightarrow$ plasmid is larger than the map $\Rightarrow$ **rearrangement/duplication** (a second copy of part of the insert, or concatenated assembly) — inferred because *fragment sum exceeds nominal size*.
- **Teaching point:** always sum fragment sizes; a size mismatch is the fingerprint of rearrangement that a single "bands match" reading can miss.
- Post-lab answers: uncut plasmid supercoiling; third-site causes; semilog ladder rationale (Module 12 §1.1).

Lab 03 — Ligation & transformation

[Instructor] Model answers & expected pattern

- Setup math: 3:1 $\Rightarrow$ 31.5 ng insert; 10:1 $\Rightarrow$ 105 ng insert (50 ng vector).
- Expected colony pattern (provided CSV):

Condition	Colonies (simulated)	Interpretation
L1 vector-only, dephosphorylated	~12	background baseline
L2 3:1, dephosphorylated	~180	good yield; screen ~8–12
L3 3:1, NOT dephosphorylated	~140 but ~85% empty	self-ligation dominated
L4 10:1, dephosphorylated	~150 but concatemer-heavy	insert excess pathology

- Screening decision: white-colony pick $\times 12$ (vector has $\text{lacZ}\alpha$) + colony PCR Set A.
- **Grading emphasis:** students must articulate that colony *number* alone can rank conditions wrongly (L3 high count, bad outcome) — screening rates are the meaningful metric.

Lab 04 — Colony PCR & screening

[Instructor] Model answers

- Set A (flanking): empty $\approx$ 0.20 kb; insert $\approx$ 1.05 kb; orientation-blind.
- Set B (vector-F + insert-R): correct orientation $\approx$ 0.7 kb; wrong orientation = no band.
- **CSV ground truth (24 colonies):** 14 correct; 4 empty; 2 wrong-orientation; 2 mixed; 2 PCR-failure.
- Blue-white cross-check: all 4 empty colonies blue (consistent); 1 blue colony is PCR-positive $\rightarrow$ carry forward for digest (possible $\text{lacZ}\alpha$ -sparing insertion at MCS edge) — designed teaching anomaly.
- Sanger plan: F + R vector-primed reads (≈ 700 –900 bp each) cover 850 bp insert + junctions; no internal primer needed.
- **Grading emphasis:** the "PCR-positive but digest-negative" contradiction (one mixed colony) — students must resolve it by trusting *structural* evidence over *amplification* evidence.

Lab 05 — qPCR expression analysis

[Instructor] Model answers

- Reference selection: *ef1a* M = 0.31, *rpl13a* M = 0.34 (stable); use geometric mean.
- Worked *myod1* @ 18 hpf (simulated data): mean Cq target 24.1; mean Cq ref 15.5 $\rightarrow \Delta\text{Ct}$ 8.6; calibrator (10 hpf) ΔCt 11.5 $\rightarrow \Delta\Delta\text{Ct} = -2.9 \rightarrow \text{FC} \approx 7.5\times$.

- Stage effect: *myod1* ANOVA $p < 0.001$ (12→18 hpf rise); *shha* $p = 0.21$ (stable); *pax6a* $p = 0.004$ (rise then plateau).
- Reporting-standard check: $n = 4$ biological; stats on ΔCt ; fold change reported with CI or SD of ΔCt .
- Spatial validation expectation: *myod1* signal in somites/adaxial cells by ISH at 18 hpf (Lab 07 panels).

Lab 06 — Reporter analysis

[Instructor] Model answers

- Normalization: ratio Firefly/Renilla per replicate → mean $\pm$ SD ($n = 4$).
- Simulated results: WT promoter 12.3 ± 1.1 ; motif-mutant 2.4 ± 0.5 (~80% reduction); minimal-only 1.1 ± 0.2 ; enhancer-A 9.8 ± 0.9 ; enhancer-B 9.1 ± 1.0 (orientation-flexible ⇒ enhancer-like); empty 1.0 ± 0.1 .
- Statistics: t-test WT vs mutant $p < 0.001$; WT vs minimal $p < 0.001$; A vs B $p = 0.38$ (n.s. → consistent with enhancer behavior).
- Fluorescence panel: notochord GFP ■ somite $\approx$ neural-tube background — concordant with endogenous *shha* literature.
- Validity ceiling (must appear in student answers): reporter $\neq$ endogenous claim; sufficiency-not-necessity for enhancer.

Lab 07 — Spatial expression

[Instructor] Model answers

- Scoring key (0–3):

Panel	notochord	floor plate	somites	PSM	forebrain	eye	type
<i>shha</i>	3	3	0	0	0	0	graded DV decrease
<i>myod1</i>	0	0	3	1	0	0	domains, stage-dependent
<i>pax6a</i>	0	0	0	0	3	3	broad anterior domain
<i>ntl</i>	3	1	0	3	0	0	staging marker
sense / no-probe	≤ 1 everywhere						background

- Gradient vs domain: *shha* DV profile decreases continuously (monotonic, no plateau-to-zero step) ⇒ gradient-consistent; students must state the quantitative criterion they used.
- Staging check: *ntl* PSM positivity confirms pre-mid-somitogenesis for the *myod1* panel.
- Probe design (*egr2b/krox20*): hindbrain rhombomeres 3/5 expected; same control set.

Lab 08 — Computational expression

[Instructor] Model answers

- **Bulk:** 18 genes at $FDR < 0.05$ & $|\log_2 FC| > 1$; top hit $\log_2 FC = -3.2$ (KO-down); volcano separates a coherent 5-gene module (provided pathway list) — discuss batch balance (metadata confirms 3 WT/3 KO balanced across 2 batches).
- **Single-cell:** QC keeps ~540/600 cells; $k=3$ clusters ≈ 220 (*pax6a*-high, neural) / 210 (*myod1*-high, muscle) / 170 (*hbbe1*-high, blood); $k=4$ reveals a 60-cell intermediate (low *myod1*, low *actc1*) ⇒ differentiating muscle precursors.

- **Spatial:** region markers recover annotation labels (spot-level accuracy ~92%); edge-gradient gene $r = -0.6$ (expression highest at tissue edge, $p < 0.001$, $n = 200$ spots).
- **Teaching emphasis:** each analysis ends with a validation plan (ISH for the gradient gene; qPCR for the top DE gene).

Lab 09 — Capstone

[Instructor] Rubric (100 pts)

Component	Points	Excellence marker
Question & design	25	falsifiable, single-variable, resolution matched to question
Execution / analysis	25	correct workflows; QC explicit; stats appropriate
Interpretation & validity	25	claims bounded; "not proven" list present
Defense & synthesis	15	answers <i>why</i> , not just <i>what</i> ; connects modules
Reporting standards	10	n/normalization/stats/effect-size stated; no fabricated citations

Model spine (Track A example): *shha* promoter-GFP (Lab-01 construct) + CRISPRi enhancer-test vector; EcoRI/HindIII plan; validation = colony PCR → digest → Sanger; expression readout = simulated GFP panels (Lab 06/07 logic); falsification test = motif-mutant reporter + CRISPRi endogenous enhancer with ISH readout.

Grading rubrics and reporting standards

Standard lab rubric (20 pts each):

Component	Points
Objectives met / correct workflow	6
Data tables complete & correct	5
Calculations shown	4
Interpretation (incl. controls discussion)	3
Post-lab & viva performance	2

Course reporting standards (apply everywhere):

1. **n** = biological replicates (technical replicates stated separately).
2. **Normalization:** reference genes / size factors named.
3. **Statistics:** test + correction + threshold.
4. **Effect size:** fold change / log2FC, not only p-values.
5. **Validation:** every major claim paired with its independent-method check.
6. **Integrity:** simulated data labeled as simulated; no fabricated citations or results.